

Homework 2 Friday, February 20, 2026, 6pm (via Gradescope.com)**Problem 1** Consider the linear time-varying system $\dot{x} = A(t)x$ with

$$A(t) := \begin{bmatrix} -1 + 1.5 \cos^2(t) & 1 - 1.5 \sin(t) \cos(t) \\ -1 - 1.5 \sin(t) \cos(t) & -1 + 1.5 \sin^2(t) \end{bmatrix}$$

1. Show that $A(t)$ is Hurwitz for each $t \geq 0$.
2. Show that the solution starting from $(1, 0)^T$ at time $t_0 = 0$ is

$$\begin{bmatrix} e^{0.5t} \cos(t) \\ -e^{0.5t} \sin(t) \end{bmatrix}.$$

3. What can you conclude about trying to deduce asymptotic stability of the origin of $\dot{x} = A(t)x$ from the eigenvalues of $A(t)$ for each $t \geq 0$?

Problem 2 Find the explicit solution of the linear time-varying system $\dot{x} = (6t \sin(t) - 2t)x$ as a function of (t, t_0, x_0) . Explain why $\lim_{t \rightarrow \infty} x(t) = 0$ for each (t_0, x_0) . Use the sequence of starting times $t_0 = 2n\pi$, $n \in \mathbb{Z}_{\geq 0}$, to show that the origin is not (uniformly) stable.**Problem 3** Use Matrosov's theorem to show that the origin of

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\sin(x_1) - g(t)x_2 \end{aligned}$$

is uniformly locally asymptotically stable when there exist $0 < a < b$ such that $g(t) \in [a, b]$ for all $t \geq 0$.**Problem 4** Find a Lyapunov function that establishes uniform global exponential stability of the origin for the system

$$\dot{x} = \left(-1 + 2 \cos(t) \right) x$$

Hint: Consider a function of the form

$$V(x, t) := \frac{1}{2} \exp \left(a \sin(t) + b \cos(t) \right) x^2.$$

Problem 5 Consider the system in Problem 4.5 of the text, i.e., $\dot{x} = g(t)f(x)$ where the origin of $\dot{x} = f(x)$ is GAS, but assume that there exists $\mu > 0$ such that $g(t) \in [0, \mu]$ for all $t \geq 0$ and $g(\cdot)$ is persistently exciting. Show that the origin is UGAS using Matrosov's theorem.**Problem 6** Consider the system in Problem 4.6 of the text:

$$\begin{aligned} \dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= ax_1 - ax_2 - g(t)x_2^3 \end{aligned}$$

where $a > 0$ (Note that, from what I have heard, the international, paperback version of the book has typos in the equations of motion). Instead of using the assumption on g given in Problem 4.6 of the text, use the following assumption: $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is bounded and persistently exciting. Show that, under this assumption, the origin of the system is uniformly globally asymptotically stable.

Problem 7 *Text problem 4.14(6),(10).*

Problem 8 Establish that the following system is ISS with linear gain with respect to the input u for the system

$$\dot{x}_1 = -x_1 + g(t)x_2 \quad (1a)$$

$$\dot{x}_2 = -g(t)x_1 + g(t)x_3 \quad (1b)$$

$$\dot{x}_3 = -g(t)x_2 + u \quad (1c)$$

under the assumption that $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is continuously differentiable and bounded with a bounded derivative and $t \mapsto g^2(t)$ is persistently exciting. Hints: 1) Use Matrosov's theorem to establish that the origin of $\dot{x} = A(t)x$ is UGAS, 2) use that if the origin of $\dot{x} = A(t)x$, where $x \in \mathbb{R}^n$, is UGAS and $A(\cdot)$ is continuous and bounded then there exist $c_1 > 0$, $c_2 > 0$, and a C^1 function $P : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ such that, for all $t \geq 0$,

$$c_1 I \leq P(t) = P(t)^T \leq c_2 I$$

and

$$\dot{P}(t) + A(t)^T P(t) + P(t) A(t) \leq -I.$$

3) Assume that the result of Theorem 4.6 holds for time-varying systems and time-varying Lyapunov functions (it does) and use $V(x, t) = x^T P(t)x$ to show ISS with linear gain for the system $\dot{x} = A(t)x + Bu$.